Exp-1
16/9/26

Basic Networking Commands

arp -a : Aim : To execute basic networking commands in Linux and window operating system.

arp -a : ARP is short form of address resolution protocol and MAC address of your system.

hostname : This is the simplest of all TCP/IP commands. It simply displays the computer.

Ipconfig/all : detail configuration information about including Router, Gateway, DNS, DHCP, and Others.

nbstat -a : (Uppercase statistics) it displays a variety of statistics about a computer TCP/IP connection.

nbstat -a : This command lists some known with NetBIOS. name resolution TCP/IP)

Pathping : Pathping traces the route to the destination address then send test. of each.

Ping : (Packet INET Groper) command is the best Ping use ICMP.

Route : Route command is used to show / manipulate the IP routing table.

```
C:\Users\DELL>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list     Loose source route along host-list.
    -h maximum_hops  Maximum number of hops to search for target.
    -i address       Use the specified source address.
    -n               Do not resolve addresses to hostnames.
    -p period        Wait period milliseconds between pings.
    -q num_queries   Number of queries per hop.
    -w timeout       Wait timeout milliseconds for each reply.
    -4               Force using IPv4.
    -6               Force using IPv6.
```

```
PS C:\Users\DELL> ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R             Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr     Source address to use.
    -c compartment Routing compartment identifier.
    -p             Ping a Hyper-V Network Virtualization provider address.
    -4             Force using IPv4.
    -6             Force using IPv6.
```

```
PS C:\Users\DELL> hostname
HDCP1126182
PS C:\Users\DELL> ipconfig / all

Error: unrecognized or incomplete command line.

USAGE:
    ipconfig [/allcompartments] [/? | /all |
                                 /renew [adapter] | /release [adapter] |
                                 /renew6 [adapter] | /release6 [adapter] |
                                 /flushdns | /displaydns | /registerdns |
                                 /showclassid adapter |
                                 /setclassid adapter [classid] |
                                 /showclassid6 adapter |
                                 /setclassid6 adapter [classid] ]

where
    adapter             Connection name
                        (wildcard characters * and ? allowed, see examples)

    Options:
       /?               Display this help message
       /all             Display full configuration information.
       /release         Release the IPv4 address for the specified adapter.
       /release6        Release the IPv6 address for the specified adapter.
       /renew           Renew the IPv4 address for the specified adapter.
       /renew6          Renew the IPv6 address for the specified adapter.
       /flushdns        Purges the DNS Resolver cache.
       /registerdns     Refreshes all DHCP leases and re-registers DNS names
       /displaydns      Display the contents of the DNS Resolver Cache.
       /showclassid     Displays all the dhcp class IDs allowed for adapter.
       /setclassid      Modifies the dhcp class id.
       /showclassid6    Displays all the IPv6 DHCP class IDs allowed for adapter.
       /setclassid6     Modifies the IPv6 DHCP class id.


The default is to display only the IP address, subnet mask and
default gateway for each adapter bound to TCP/IP.

For Release and Renew, if no adapter name is specified, then the IP address
leases for all adapters bound to TCP/IP will be released or renewed.

For Setclassid and Setclassid6, if no ClassId is specified, then the ClassId is removed.

Examples:
    > ipconfig                      ... Show information
    > ipconfig /all                 ... Show detailed information
    > ipconfig /renew               ... renew all adapters
    > ipconfig /renew EL*           ... renew any connection that has its
                                        name starting with EL
    > ipconfig /release *Con*       ... release all matching connections,
                                        eg. "Wired Ethernet Connection 1" or
                                            "Wired Ethernet Connection 2"
    > ipconfig /allcompartments     ... Show information about all
                                        compartments
    > ipconfig /allcompartments /all ... Show detailed information about all
                                         compartments
PS C:\Users\DELL>
```

1b) Linux Commands:

ip : A versatile command for managing network configuration IP address and routing tables. It is used to assign or delete IPs flagged interface status.

(i) ip addr show
(ii) ip addr del 192.168.1.254/24 dev eth0
(iii) ip link set eth0 promisc on
(iv) ip route add default via 192.16.1.254 dev eth0.
(v) ip link set eth0 up.
(vi) ip link set eth0 promisc on
(vi) ip route add default

(2) ipconfig : A stage tool historically used for config and troubleshooting network interface. It has largely been and troubleshooting network interface. It has largely been replaced by the ip command.

(3) Mtr : Combines the functioning of ping and traceroute to provide diagnostic statistics for each network hop.

* mtr google.com
* mtr -4 google.com
* mtr -6 google.com
* mtr -c google.com.



```
PS C:\Users\DELL> nslookup
Default Server:  Unknown
Address:  172.16.72.1
```



